

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
Student Name:	Athavan K	Reg. No.:	192524036
Section / Batch:	AI&DS	Date:	

Instructions to Students

- Answer the following question. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type	Focus Area	Questions
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1

SECTION A – APPLICATION BASED QUESTIONS**Q1) Searching Techniques – Efficiency Comparison on Sorted Dataset**

Given the sorted dataset [4, 8, 12, 16, 20, 24, 28, 32, 36], analyze the performance of Linear Search and Binary Search to find the element 28. Clearly interpret the problem and explain the working of both searching methods. Show the step-by-step procedure for each algorithm and count the number of comparisons required. Analyze the time complexity of both approaches. Compare their efficiency for sorted data. Interpret the results and justify which method is preferable in practical applications.

Code of Conduct

I Athavan K certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Athavan K

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%				
Application of Concepts	30%				
Problem-Solving Approach	20%				
Accuracy of Solution	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

C02 - AT-1

Analytical Questions

Name: Athavan K

Reg no: 192524036

Course code: CSD0614

Course Design & analysis
of algorithm

1) Searching Techniques - Efficiency Comparison on Sorted Dataset

Problem Interpretation

Given the Sorted dataset $[4, 8, 12, 16, 20, 24, 28, 32, 36]$, the Objective is to search for the element 28 using Linear Search and Binary Search. The performance of both algorithm is analyzed by showing their working steps, counting the complexity and comparing their efficiency on sorted data.

Linear Search

Linear Search checks each element one by one from the beginning until the required element is found.

Steps:

- 1) Compare 28 with 4 $\rightarrow$ NOT Found (1 comparison)
- 2) Compare 28 with 8 $\rightarrow$ NOT Found (2 comparison)
- 3) Compare 28 with 12 $\rightarrow$ NOT Found (3 comparisons)
- 4) Compare 28 with 16 $\rightarrow$ NOT Found (4 comparisons)
- 5) Compare 28 with 20 $\rightarrow$ NOT Found (6 comparisons)

0) Compare 28 with 24 $\rightarrow$ NOT Found (6 Comparisons)

1) Compare 28 with 28 $\rightarrow$ Found (7 Comparisons)

Total Comparisons: 7

Time Complexity

* Best Case: $O(1)$

* Average Case: $O(n)$

* Worst Case: $O(n)$

2) Binary Search

Binary Search works only on sorted data. It repeatedly divides the search space into two halves until the element is found.

Steps

1) Middle Element = 20 $\rightarrow$ 28 > 20, Search right half (1 - Comparison)

2) Right half = [24, 28, 32, 36], middle = 28 $\rightarrow$ Found (2 Comparisons)

Total Comparison: 2

Time Complexity

* Best Case: $O(1)$

* Average Case: $O(\log n)$

* Worst Case: $O(\log n)$